

2 A student places a slide with a double slit on a support clamped to the bench as shown in Fig. 2.1.

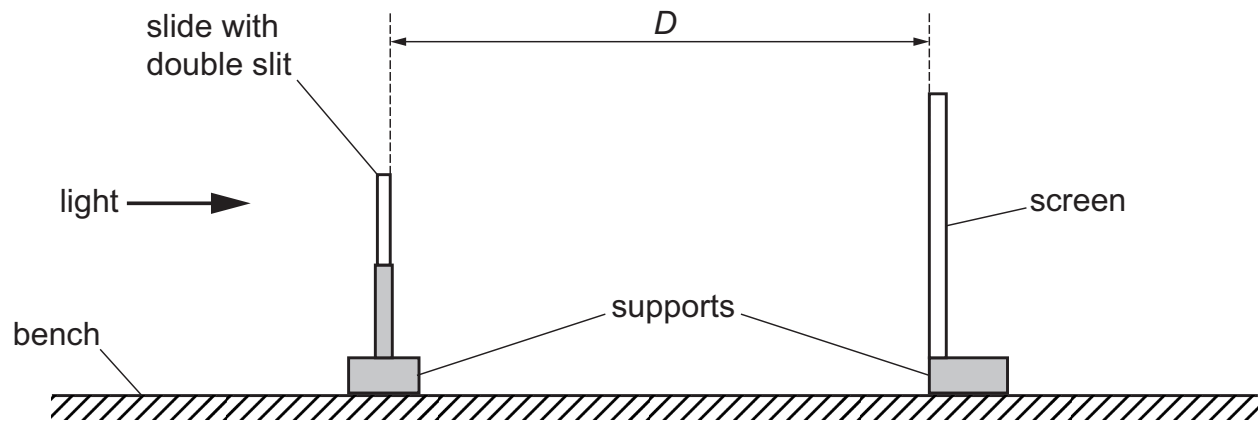


Fig. 2.1

The distance between the slide and the screen is D .

The separation s of the slits is determined.

Light from a laser is incident normally on the double slit. An interference pattern is observed on the screen. The distance w across 10 fringes is measured. The distance y between the centres of adjacent fringes is calculated using the equation

$$y = \frac{w}{10}.$$

The experiment is repeated with slides of different slit separation s .

It is suggested that y and s are related by the equation

$$\lambda = \frac{sy}{D}$$

where λ is the wavelength of the incident light.

(a) A graph is plotted of y on the y -axis against $\frac{1}{s}$ on the x -axis.

Determine an expression for the gradient.

gradient = [1]

(b) Values of s and w are given in Table 2.1.

Table 2.1

s/mm	$\frac{1}{s}/\text{mm}^{-1}$	w/mm	y/mm
0.18 ± 0.01		33.0	
0.21 ± 0.01		28.9	
0.24 ± 0.01		25.1	
0.27 ± 0.01		22.6	
0.31 ± 0.01		19.6	
0.38 ± 0.01		15.9	

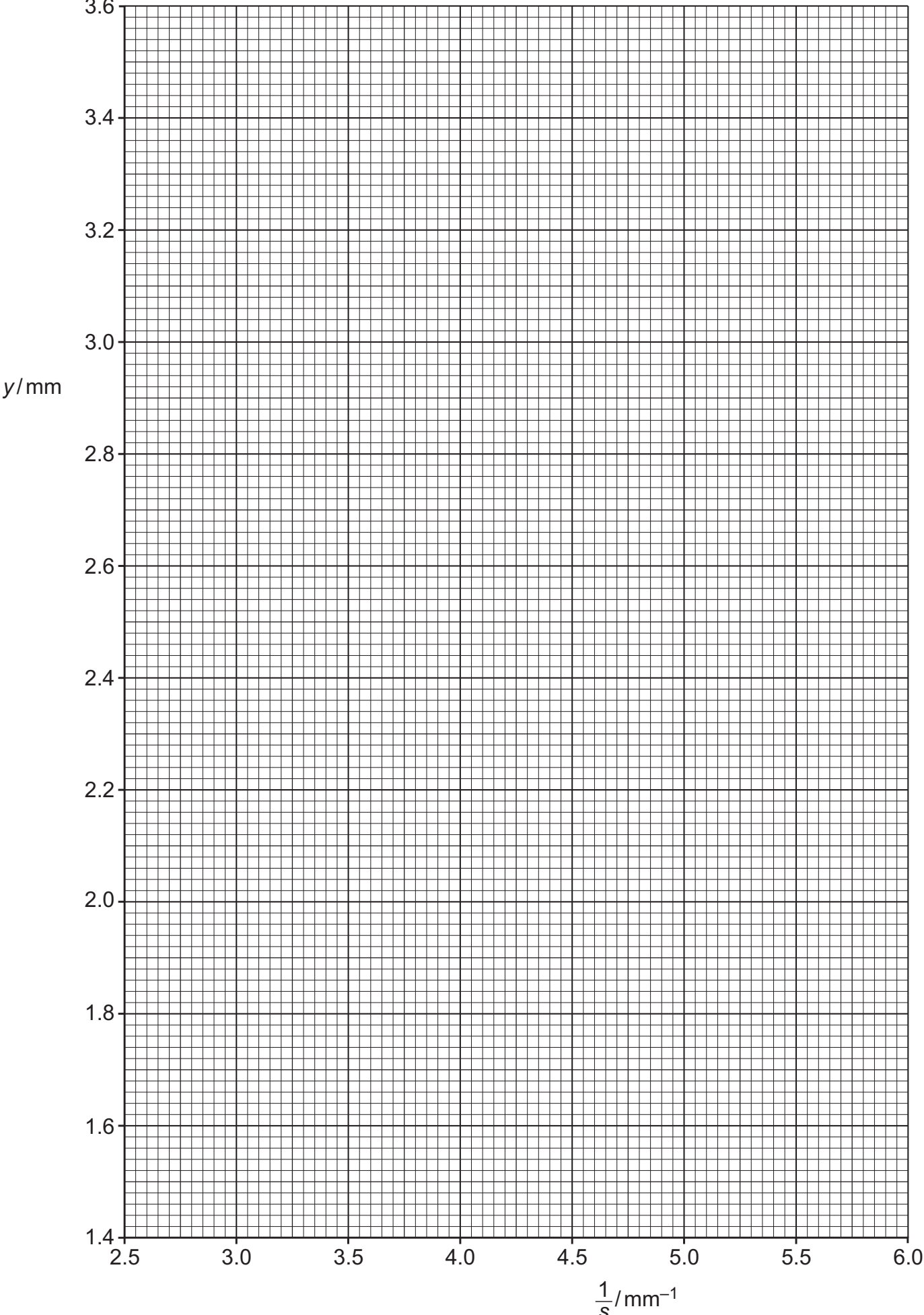
Calculate and record values of $\frac{1}{s}/\text{mm}^{-1}$ and y/mm in Table 2.1. Include the absolute uncertainties in $\frac{1}{s}$. [2]

(c) (i) Plot a graph of y/mm against $\frac{1}{s}/\text{mm}^{-1}$. Include error bars for $\frac{1}{s}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



(d) The distance between the slide and the screen is measured several times:

0.929 m 0.918 m 0.913 m 0.918 m 0.927 m.

Determine the mean distance D . Include the absolute uncertainty.

$D = \dots\dots\dots\text{m}$ [1]

(e) (i) Using your answers to (a), (c)(iii) and (d), determine the value of λ . Include an appropriate unit.

$\lambda = \dots\dots\dots$ [2]

(ii) Determine the percentage uncertainty in λ .

percentage uncertainty in $\lambda = \dots\dots\dots\%$ [1]

(f) The experiment is repeated. Determine the slit separation s that gives a value of y of $(0.500 \pm 0.005)\text{cm}$. Include the absolute uncertainty.

$s = \dots\dots\dots\text{m}$ [2]